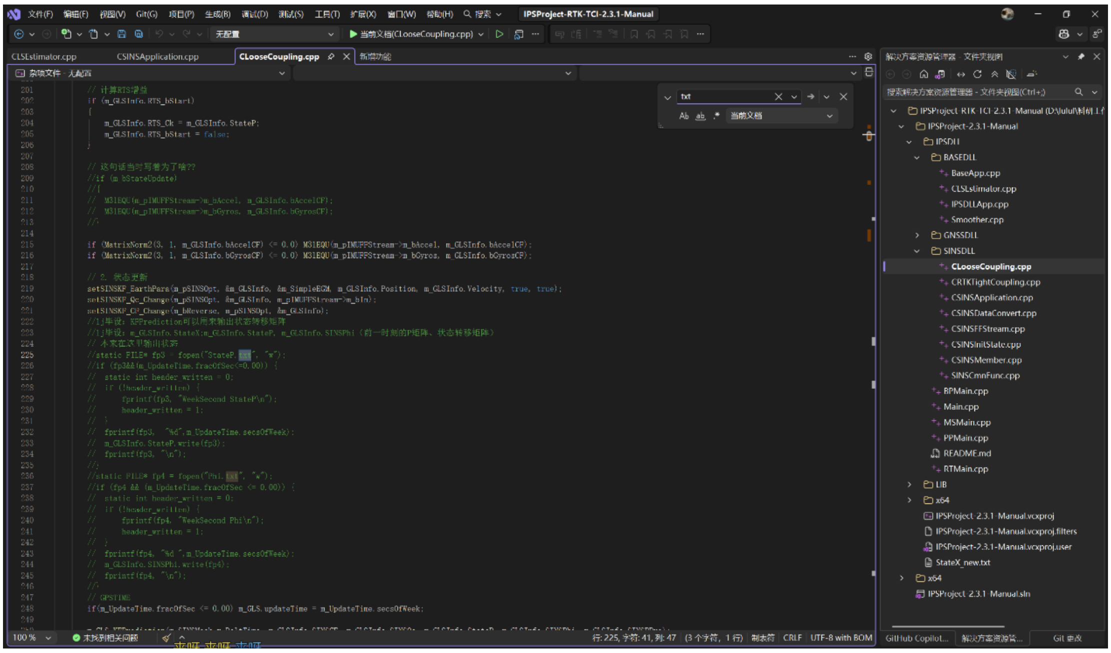
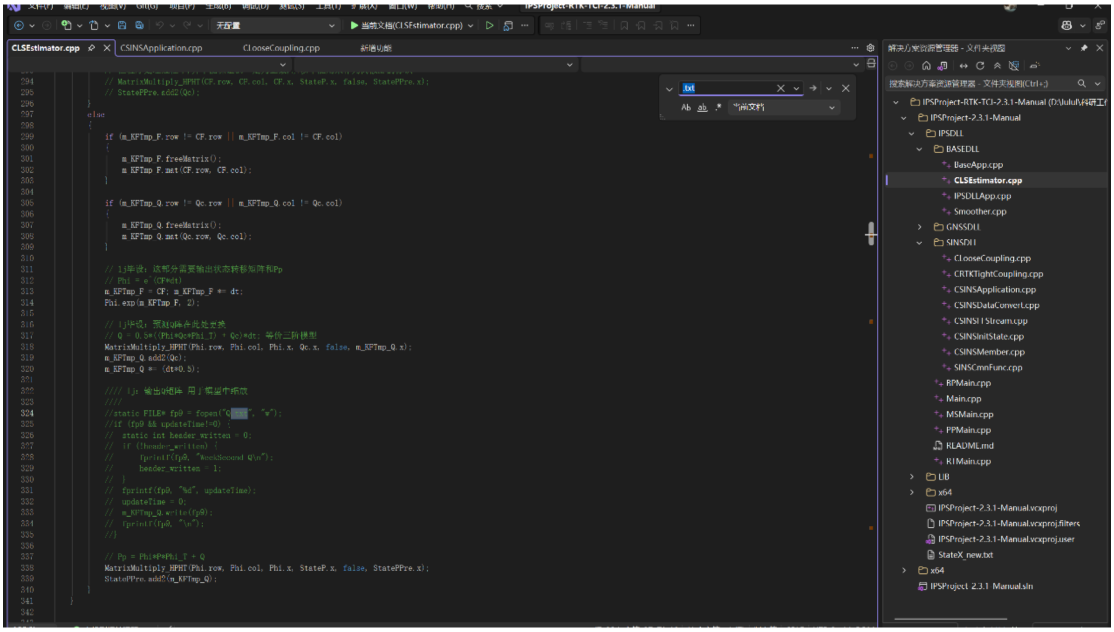
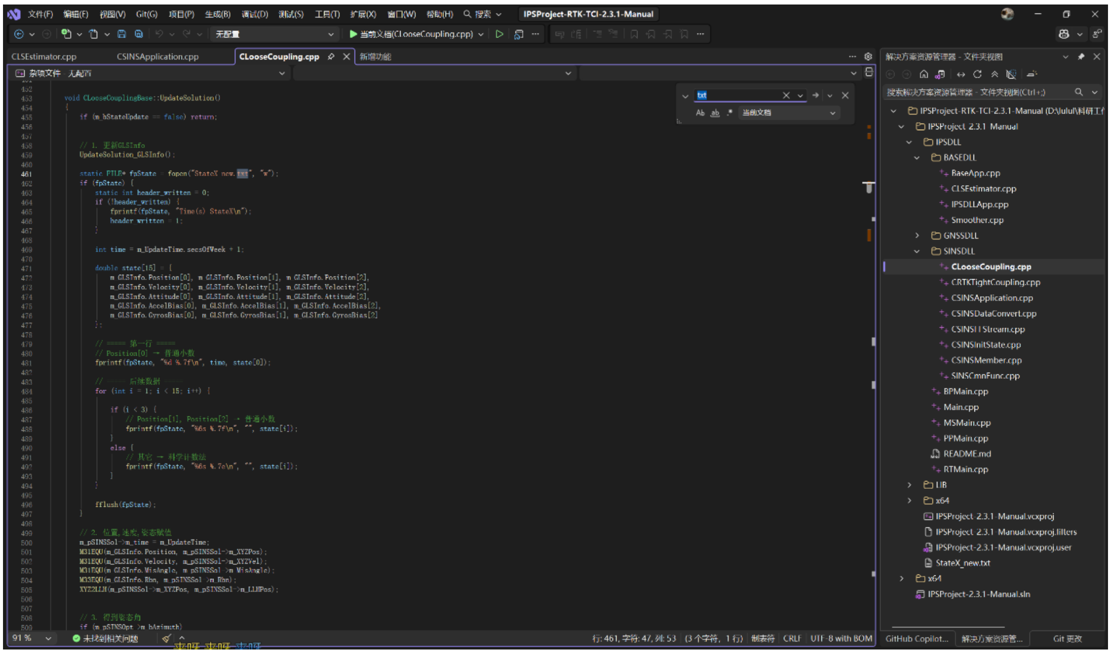


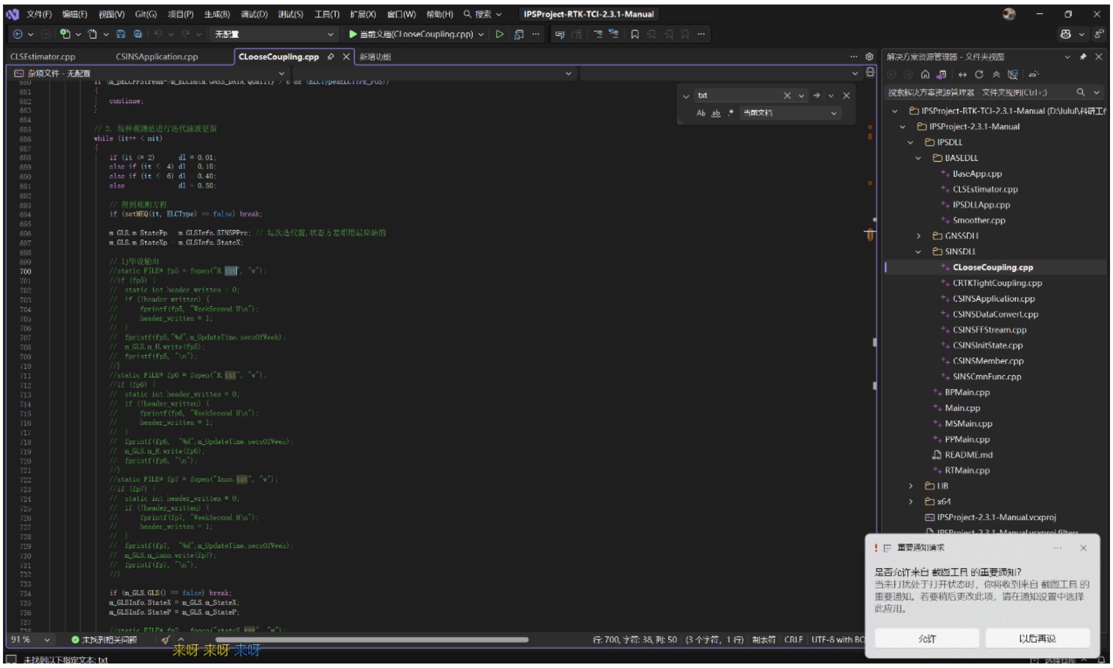
```
// 计算RTS增益
if (m_GLSInfo.RTS_bStart)
{
    m_GLSInfo.RTS_Ck = m_GLSInfo.StateP;
    m_GLSInfo.RTS_bStart = false;
}

// 这句话当时写着为了啥??
//if (m_bStateUpdate)
//{
//  M31EQU(m_pIMUFFStream->m_bAccel, m_GLSInfo.bAccelCF);
//  M31EQU(m_pIMUFFStream->m_bGyros, m_GLSInfo.bGyrosCF);
//}

if (MatrixNorm2(3, 1, m_GLSInfo.bAccelCF) <= 0.0) M31EQU(m_pIMUFFStream->m_bAccel, m_GLSInfo.bAccelCF);
if (MatrixNorm2(3, 1, m_GLSInfo.bGyrosCF) <= 0.0) M31EQU(m_pIMUFFStream->m_bGyros, m_GLSInfo.bGyrosCF);

// 2. 状态更新
setSINSKF_EarthPara(m_pSINSOpt, &m_GLSInfo, &m_SimpleEGM, m_GLSInfo.Position, m_GLSInfo.Velocity, true, true);
setSINSKF_Qc_Change(m_pSINSOpt, &m_GLSInfo, m_pIMUFFStream->m_bIn);
setSINSKF_CF_Change(m_bReverse, m_pSINSOpt, &m_GLSInfo);
//lj毕设：KFPrediction可以用来输出状态转移矩阵
//lj毕设：m_GLSInfo.StateX;m_GLSInfo.StateP, m_GLSInfo.SINSPhi（前一时刻的P矩阵、状态转移矩阵）
// 本来在这里输出状态
//static FILE* fp3 = fopen("StateP.txt", "w");
//if (fp3&&(m_UpdateTime.fracOfSec<=0.00)) {
//  static int header_written = 0;
//  if (!header_written) {
//      fprintf(fp3, "WeekSecond StateP\n");
//      header_written = 1;
//  }
//  fprintf(fp3, "%d",m_UpdateTime.secsOfWeek);
//  m_GLSInfo.StateP.write(fp3);
//  fprintf(fp3, "\n");
//}
//static FILE* fp4 = fopen("Phi.txt", "w");
//if (fp4 && (m_UpdateTime.fracOfSec <= 0.00)) {
//  static int header_written = 0;
//  if (!header_written) {
//      fprintf(fp4, "WeekSecond Phi\n");
//      header_written = 1;
//  }
//  fprintf(fp4, "%d ",m_UpdateTime.secsOfWeek);
//  m_GLSInfo.SINSPhi.write(fp4);
//  fprintf(fp4, "\n");
//}
// GPSTIME
if(m_UpdateTime.fracOfSec <= 0.00) m_GLS.updateTime = m_UpdateTime.secsOfWeek;
```



```
    // MatrixMultiply_HPHT(CF.row, CF.col, CF.x, StateP.x, false, StatePPre.x);
    // StatePPre.add2(Qc);
}
else
{
    if (m_KFTmp_F.row != CF.row || m_KFTmp_F.col != CF.col)
    {
        m_KFTmp_F.freeMatrix();
        m_KFTmp_F.mat(CF.row, CF.col);
    }

    if (m_KFTmp_Q.row != Qc.row || m_KFTmp_Q.col != Qc.col)
    {
        m_KFTmp_Q.freeMatrix();
        m_KFTmp_Q.mat(Qc.row, Qc.col);
    }

    // lj毕设：这部分需要输出状态转移矩阵和Pp
    // Phi = e^(CF*dt)
    m_KFTmp_F = CF; m_KFTmp_F *= dt;
    Phi.exp(m_KFTmp_F, 2);

    // lj毕设：预测Q阵在此处更换
    // Q = 0.5*((Phi*Qc*Phi_T) + Qc)*dt; 等价三阶模型
    MatrixMultiply_HPHT(Phi.row, Phi.col, Phi.x, Qc.x, false, m_KFTmp_Q.x);
    m_KFTmp_Q.add2(Qc);
    m_KFTmp_Q *= (dt*0.5);

    //// lj: 输出Q矩阵 用于模型中缩放
    ////
    //static FILE* fp9 = fopen("Q.txt", "w");
    //if (fp9 && updateTime!=0) {
    //  static int header_written = 0;
    //  if (!header_written) {
    //      fprintf(fp9, "WeekSecond Q\n");
    //      header_written = 1;
    //  }
    //  fprintf(fp9, "%d", updateTime);
    //  updateTime = 0;
    //  m_KFTmp_Q.write(fp9);
    //  fprintf(fp9, "\n");
    //}

    // Pp = Phi*P*Phi_T + Q
    MatrixMultiply_HPHT(Phi.row, Phi.col, Phi.x, StateP.x, false, StatePPre.x);
    StatePPre.add2(m_KFTmp_Q);
}
}
```



```cpp
void CLooseCouplingBase::UpdateSolution()
{
    if (m_bStateUpdate == false) return;


    // 1. 更新GLSInfo
    UpdateSolution_GLSInfo();

    static FILE* fpState = fopen("StateX_new.txt", "w");
    if (fpState) {
        static int header_written = 0;
        if (!header_written) {
            fprintf(fpState, "Time(s) StateX\n");
            header_written = 1;
        }

        int time = m_UpdateTime.secsOfWeek + 1;

        double state[15] = {
            m_GLSInfo.Position[0], m_GLSInfo.Position[1], m_GLSInfo.Position[2],
            m_GLSInfo.Velocity[0], m_GLSInfo.Velocity[1], m_GLSInfo.Velocity[2],
            m_GLSInfo.Attitude[0], m_GLSInfo.Attitude[1], m_GLSInfo.Attitude[2],
            m_GLSInfo.AccelBias[0], m_GLSInfo.AccelBias[1], m_GLSInfo.AccelBias[2],
            m_GLSInfo.GyrosBias[0], m_GLSInfo.GyrosBias[1], m_GLSInfo.GyrosBias[2]
        };

        // ==== 第一行 ====
        // Position[0] → 普通小数
        fprintf(fpState, "%d %.7f\n", time, state[0]);

        // ---- 后续数据 ----
        for (int i = 1; i < 15; i++) {

            if (i < 3) {
                // Position[1], Position[2] → 普通小数
                fprintf(fpState, "%6s %.7f\n", "", state[i]);
            }
            else {
                // 其它 → 科学计数法
                fprintf(fpState, "%6s %.7e\n", "", state[i]);
            }
        }

        fflush(fpState);
    }

    // 2. 位置,速度,姿态赋值
    m_pSINSSol->m_time = m_UpdateTime;
    M31EQU(m_GLSInfo.Position, m_pSINSSol->m_XYZPos);
    M31EQU(m_GLSInfo.Velocity, m_pSINSSol->m_XYZVel);
    M31EQU(m_GLSInfo.MisAngle, m_pSINSSol->m_MisAngle);
    M33EQU(m_GLSInfo.Rbn, m_pSINSSol->m_Rbn);
    XYZ2LLH(m_pSINSSol->m_XYZPos, m_pSINSSol->m_LLHPos);


    // 3. 得到姿态角
    if (m_pSINSOpt->m_bAzimuth)
```



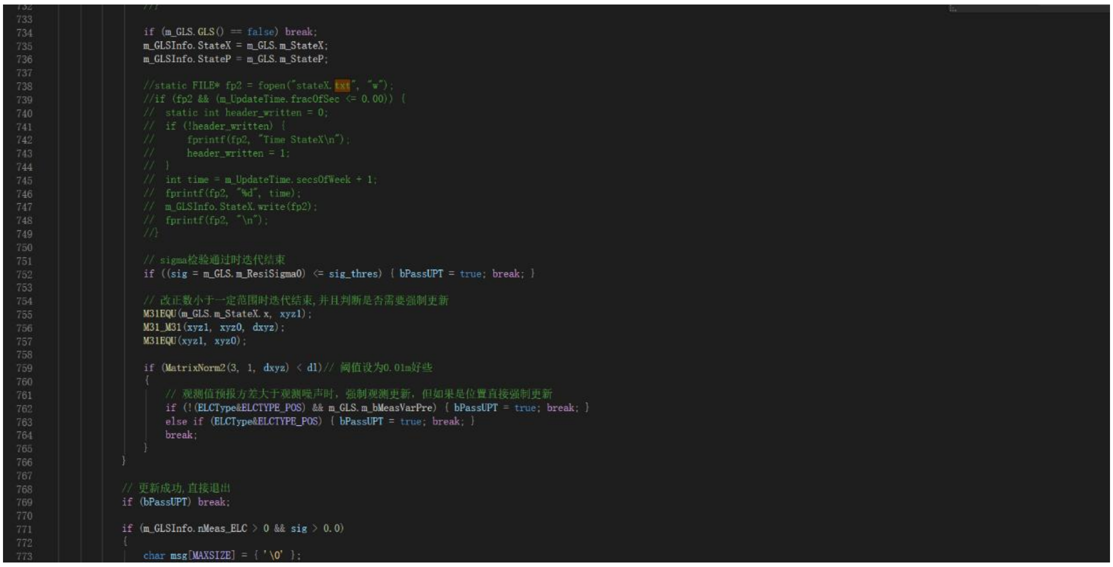
```cpp
            {
                continue;
            }

            // 2. 每种观测值进行迭代滤波更新
            while (it++ < nit)
            {
                if (it <= 2)      dl = 0.01;
                else if (it <  4) dl = 0.10;
                else if (it <  6) dl = 0.40;
                else              dl = 0.50;

                // 得到观测方程
                if (setMEQ(it, ELCType) == false) break;

                m_GLS.m_StatePp = m_GLSInfo.SINSPPre; // 每次迭代前,状态方差都用最原始的
                m_GLS.m_StateXp = m_GLSInfo.StateX;

                // 1j毕设输出
                //static FILE* fp5 = fopen("H.txt", "w");
                //if (fp5) {
                //  static int header_written = 0;
                //  if (!header_written) {
                //      fprintf(fp5, "WeekSecond H\n");
                //      header_written = 1;
                //  }
                //  fprintf(fp5,"%d",m_UpdateTime.secsOfWeek);
                //  m_GLS.m_H.write(fp5);
                //  fprintf(fp5, "\n");
                //}
                //static FILE* fp6 = fopen("R.txt", "w");
                //if (fp6) {
                //  static int header_written = 0;
                //  if (!header_written) {
                //      fprintf(fp6, "WeekSecond H\n");
                //      header_written = 1;
                //  }
                //  fprintf(fp6,  "%d",m_UpdateTime.secsOfWeek);
                //  m_GLS.m_R.write(fp6);
                //  fprintf(fp6, "\n");
                //}
                //static FILE* fp7 = fopen("Inno.txt", "w");
                //if (fp7) {
                //  static int header_written = 0;
                //  if (!header_written) {
                //      fprintf(fp7, "WeekSecond H\n");
                //      header_written = 1;
                //  }
                //  fprintf(fp7,  "%d",m_UpdateTime.secsOfWeek);
                //  m_GLS.m_Inno.write(fp7);
                //  fprintf(fp7, "\n");
                //}

                if (m_GLS.GLS() == false) break;
                m_GLSInfo.StateX = m_GLS.m_StateX;
                m_GLSInfo.StateP = m_GLS.m_StateP;
```

重要通知请求

是否允许来自 截图工具 的重要通知?

当未打扰处于打开状态时，你将收到来自 截图工具 的重要通知。若要稍后更改此项，请在通知设置中选择此应用。

允许　以后再说



```
            //}

            if (m_GLS.GLS() == false) break;
            m_GLSInfo.StateX = m_GLS.m_StateX;
            m_GLSInfo.StateP = m_GLS.m_StateP;

            //static FILE* fp2 = fopen("stateX.txt", "w");
            //if (fp2 && (m_UpdateTime.fracOfSec <= 0.00)) {
            //  static int header_written = 0;
            //  if (!header_written) {
            //      fprintf(fp2, "Time StateX\n");
            //      header_written = 1;
            //  }
            //  int time = m_UpdateTime.secsOfWeek + 1;
            //  fprintf(fp2, "%d", time);
            //  m_GLSInfo.StateX.write(fp2);
            //  fprintf(fp2, "\n");
            //}

            // sigma检验通过时迭代结束
            if ((sig = m_GLS.m_ResiSigma0) <= sig_thres) { bPassUPT = true; break; }

            // 改正数小于一定范围时迭代结束,并且判断是否需要强制更新
            M31EQU(m_GLS.m_StateX.x, xyz1);
            M31_M31(xyz1, xyz0, dxyz);
            M31EQU(xyz1, xyz0);

            if (MatrixNorm2(3, 1, dxyz) < dl)// 阈值设为0.01m好些
            {
                // 观测值预报方差大于观测噪声时，强制观测更新，但如果是位置直接强制更新
                if (!(ELCType&ELCTYPE_POS) && m_GLS.m_bMeasVarPre) { bPassUPT = true; break; }
                else if (ELCType&ELCTYPE_POS) { bPassUPT = true; break; }
                break;
            }
        }

        // 更新成功,直接退出
        if (bPassUPT) break;

        if (m_GLSInfo.nMeas_ELC > 0 && sig > 0.0)
        {
            char msg[MAXSIZE] = { '\0' };
```